Total No. of Questions : 8]

SEAT No. :

P2966

[Total No. of Pages : 5

[6004]-949

B.E. (Mechanical /Mechanical SW)

MECHANICAL SYSTEM DESIGN

(2015 Pattern) (Semester - II) (402048)

Time :3 Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) *Solve, Q.1,Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.*
2) *Figures to the right indicate ful marks.*
3) *Use of logarithmic tables slide rule, Mollier charts, electronic pocket calculator and tables is allowed.*
4) *Neat diarams must be drawn wherever necessary.*
5) *Assume suitable data, if necessary.*

Q1) a) A Six speed multispeed gearbox is to be designed for spindle speed varying from 63 rpm to 630 rpm. If the geometric progression ratio as per R5 series **[6]**

i) Draw structural Diagram (for any one Structural Formula)

ii) Draw Speed Diagram, if the gear box is to be driven by 720 rpm 3-phase AC motor through belt drive

iii) Draw gearing diagram

b) Explain the following terms used in Statistical Analysis, **[4]**

i) Mean

ii) Variance

iii) Standard Deviation Median

OR

Q2) a) Compare different laws of regulation of speeds in multispeed Gear box. **[6]**

b) The Tensile strength of a population of 700 connecting rods are normally distributed with a mean of 450 N/mm^2 and standard deviation of 50 N/mm^2 , Determine the number of connecting rods having strentgth less than 395 N/mm^2 **[4]**

Z	1	1.1	2.8	2.9
Area	0.3413	0.3643	0.4974	0.4981

P.T.O.

Q3) A Triple ply belt conveyor is required to transport 2 tons of iron ore per hour 1000m and a height of 300 m The permissible belt speed is 90 m/ min. If the mass density of iron ore is 2.5 tons per cubic meter. **[10]**

Determine

i) The bale width

ii) The diameter of drive Pulley

iii) Ratio of gear reducer if electric motor speed is 1440 rpm Use following data Flowability factor 'k'

Belt Inclination (α)	10º-15º	16º-20º	21º-25º	26º-30º	31º-35º
Flowability factor 'k'	2.65×10^{-4}	2.5×10^{-4}	2.35×10^{-4}	2.20×10^{-4}	2.05×10^{-4}

Standard belt widths :

400,450,500,600,650,750,800,900,1000,1200,1400,1600,1800,2000mm.

OR

Q4) a) The following data refers to a flat belt conveyor for transporting crushed rock: **[6]**

Mass density: 2 Tons/m^3

Belt Speed: 1.75 m/s

Belt Width: 0.8 m

Surcharge angle : 25º

Effective width of the, material carried bt the belt safely, b = (0.9 B - 0.05).

Determine the capacity of conveyor in tons/hr

b) Describe with sketch the procedure to calculate the power requirement of belt conveyors. **[4]**

Q5) a) What is auto-frettage? Describe any one method of pre-stressing the cylinders with neat sketch. **[8]**

b) A high pressure compound cylinder consists of an inner and outer diameter of 300 mm and 400 mm OD respectively. It is jacketed by an outer cylinder of 500 mm outside diameter. The tubes are assembled by a shrinking process in such a way that the maximum principal stress induced in any tube is limited to 100 N/mm^2. Calculate the shrinkage pressure and original dimensions of the tube assuming E = 210 GPa. **[10]**

OR

Q6) a) i) What are the types of end closures for cylindrical vessel. Explain the design procedure of hemispherical head. **[4]**

ii) Explain various categories of the welded joints used in unfired pressure vessel. **[4]**

b) The cylindrical pressure vessel made of steel (S_{yt} = 270 N/mm^2) is subjected to operating pressure of 2 MPa. The thickness of the shell and torispherical heads of vessel are 12 mm and 16 mm respectively. If the weld efficiency is 0.7 and corrosion allowance is 2 mm, determine **[10]**

i) Inside diameter of shell

ii) Crown radius of torispherical head

Q7) a) Explain the stresses developed in the cylinder wall **[6]**

b) The cylinder of 4 stroke diesel engine has the following specifications: **[10]**

Brake Power = 7.5 kW,

Speed = 1400 rpm,

Indicated mean effective pressure = 0.35 MPa

Mechanical Efficiency = 80% and

Maximum gas pressure = 3.5 Mpa.

The liner and cylinder head are made of gray cast iron FG250 (S_{ut} = 250 MPa and v = 0.25). If the factor of safety for all parts is 6,

calculate:

i) the bore and length of cylinder liner;

ii) the thickness of cylinder liner;

iii) the thickness of cylinder head.

OR

Q8) The following data refers to a single cylinder four stroke diesel engine:**[16]**

a) Cylinder bore = 100 mm,

b) Piston stroke = 125 mm,

c) Speed = 2000 rpm,

d) Brake mean effective pressure = 0.65 N/mm^2,

e) Maximum gas pressure = 5 N/mm^2,

f) Brake specific fuel consumption = 0.25kg/kW-h

g) Higher calorific value of fuel = 42,000 kj/kg

h) Thermal conductivity of cast iron piston = 46.5 W/mºC,

i) Allowable temperature difference for piston = 220ºC,

j) Heat conducted through piston crown = 5% of heat generated during combustion,

k) Radial pressure between piston ring and cylinder wall = 0.04 N/mm^2,

l) Permissible bearing pressure between piston skirt and cylinder wall = 0.04 N/mm^2,

m) Permissible bearing pressure between alloy steel piston pin and bronze bush in connecting rod = 25 N/mm^2,

n) Allowable tensile stress for C.I. Piston = 37.5 N/mm^2,

o) Allowable bending stress for C.I.alloy piston rings = 100 N/mm^2,

p) Allowable shear stress alloy steel piston pin = 80 N/mm^2,

q) Allowable bending stress for alloy steel piston pin = 160 N/mm^2.

Design:

i) Piston Head

ii) Piston barrel

iii) Piston rings

iv) Piston Skirt

v) Reinforcing ribs

vi) Piston pin

Q9) a) What is optimum design? Explain with examples **[6]**

b) A Helical compression spring to be used in exhaust valve mechanism is initially compressed with a preload of 500 N and the valve lift is 40 mm. Design the spring with modulus of rigidity 90 GPa and Wahl's shear stress factor as 1.14 such that the torsional shear stress in spring will not exceed 700 MPa. The spring would weight minimum with the condition that outside diameter fixed at 60 mm when optimized. std Wire dia: 4,4.5,5,5.6,6,6.5,7,8,8.5,9,9.5,10 mm **[10]**

OR

Q10) A shaft is to be used to transmit a torque of 900 N-m. The required torsional stiffness is 90 N-m / degree while the factor of safety based on yield strength is 1.5. Using maximum shear stress theroty, design the shaft with the objective of minimizing the weight. What will be the change in design for minimum cost? **[16]**

Material	Mass Density ρ,kg/mm^3	Modulus of rigidity G,(GPa)	Yield Strength S_{yt},MPa	Material Cost, C (Rs/N)
MAT 1	8500	80	130	16
MAT 2	3000	26.5	50	32
MAT 3	4800	40	90	480
MAT 4	2100	16	20	32